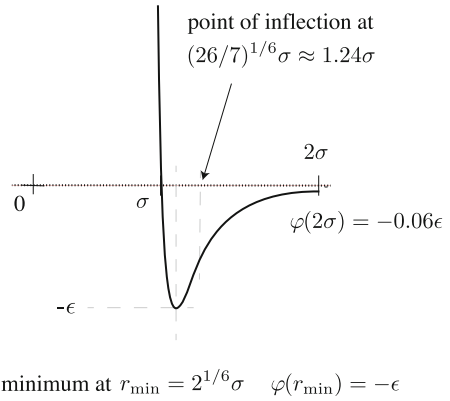


Fig. 1.6 The Lennard-Jones pair potential $\varphi(r) = 4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6]$ is shown

of short-ranged soft walls in molecular dynamics, i.e., the fact that φ_{LJ} tends rapidly to positive infinity as $r \to 0$, the atoms remain well separated in long simulations. The singularity at $r = 0$ is therefore rarely encountered in dynamics trajectories, however the presence of the singularity may nonetheless create problems for mathematical analysis, as many theoretical techniques rely on assumed smoothness. An alternative would be to replace the Lennard-Jones potential with a smooth fitted (e.g. piecewise polynomial) potential that captures, for a sufficiently large range of energies, both the attractive behavior and the strong repulsion, while mimicking the Lennard-Jones potential adequately on a sufficiently large interval, without introducing a singularity.

When there are many atoms of different types, the parameters of the potential will depend on this, so we obtain contributions to the total potential energy of the form

$$\varphi_{ij}^{\text{LJ}}(r_{ij}) = 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right],$$

where $r_{ij} = \|\boldsymbol{q}_i - \boldsymbol{q}_j\|$. The Lennard-Jones potential is graphed in Fig. 1.6.

1.1.2 Other Potentials

As we have already mentioned, an important and growing application area for molecular simulation methods lies in the modelling of biological molecules. For example, the alanine dipeptide is a very popular model for examining different simulation algorithms intended for use in biomolecular modelling. It consists of 22 atoms (carbon, oxygen, hydrogen, nitrogen) arranged as shown in the ball and stick diagram of Fig. 1.7.